

ARBA-MINTCH UNIVERSITY
ARBA-MINTCH INSTITUTE OF TECHNOLOGY
CIVIL ENGINEERING DEPARTMENT

Target group: G4C-A,B	Semester: II	Maximum mark allowed: 50%
Exam type: Closed	Course title: Highway engineering-II	Course code: CEng:4183
Maximum allowed time: 3:00	Fix your calculator two digit numbers	Final exam

1. Assume that you are assigned as asphalt engineer in asphalt mix plant in Arba-Minch town which owned by DMC construction private limited company. As a civil engineer, write the procedures you follow to produce high quality asphalt mix. Besides, explains in detail the objectives your mix should attain in the design period. **(5marks)**
2. State and discuss the basic difference between maintenance and rehabilitation. **(5marks)**
3. a. State and discuss why volumes are expressed in mass in mass rather than volume. **(1.5Mark)**
b. Discuss why minimum air voids are provided in asphalt mix during road construction. **(1.5 Marks)**
c. The weight of compacted specimen is air 1167.80g, 1169.20g saturated and 670.07g when submerged in water respectively. In addition, the mixture is mixed at a ratio of 1:8:6:5 asphalt cement, coarse aggregate, fine aggregate and filler aggregate material correspondingly. Additional necessary information is tabulated below. Determine the VMA, VTM, VFA, Pba, Pbe, Gmb and Gmm of the mixed asphalt. **(7marks)**

Material	Asphalt cement	Coarse Aggregate	Fine Aggregate	Filler aggregate
Apparent specific gravity	1.00	2.61	2.71	2.75
Bulk specific gravity	1.00	2.51	2.63	2.69

4. a. Define normal traffic, diverted traffic and generated traffic as well as design period. **(2 Marks)**
b. Construction of a two lane link road to connect two towns is under proposal. Sub-grade soil soaked CBR strength on samples taken from every 0.5km along the line respectively in percent are: 8,9,10,11,9,10,11,9,10,9,11. The road is going to design with 90% reliability.

The road is expected to generate average daily traffic (ADT) in its initial service year 450vehicles per day in both directions with equal directional split. Among these, 20%

small car category, 20% bus category, 20% medium truck category, 20% heavy truck category and 20% articulated truck with damaging factors 0.00, 0.14, 6.67, 11.47 and 14.30 respectively. In addition, 20% traffics will be generated and diverted at year of opening with same traffic composition to the normal traffic. The road construction is expected to begin after one year traffic survey and open after two years of construction time. Determine the design CBR and Estimate the design 80kN ESA load of the pavement for its design period. The traffic growth rate is 5% throughout the country. You have to use the first chart in CBR determination. **(8marks)**

5. The surface water from road side with run of coefficient **0.8** is drained to the longitudinal side drain from across $\frac{1}{2}$ a bituminous pavement surface of total width **7m**, shoulder, side slope and adjoining land width **8m** with run of coefficient **0.35** on one side of the drain. In addition, the distance between the farthest point in the turf covered drainage area and the point of entry to side drain is **175m** with run-off coefficient is **0.25**. The length of the longitudinal open drain in a sandy clay soil from the inlet point to the cross drainage is **500m**. The velocity of flow in the side drain may be assumed as **1m/sec** so that silting and erosion are prevented.

Estimate the design quantity of flow on the side drain if the rainfall intensity is estimated **120mm/hr**. Design the cross section and longitudinal slope of economical **rectangular** drain assuming that Manning's roughness coefficient is 0.02. **(10 marks)**

6. Draw the trial mix results vs. bitumen content on the sheet provided and suggest the optimum bitumen content for asphalt concrete with 19mm nominal aggregate size and check whether the specification for heavy traffic area satisfied or not after drawing from chart-2 to chart-6.

Bitumen content (%)	Density (KN/m ³)	Stability (kN)	Flow (0.25mm)	VTM (%)	VMA (%)
3.5	23	8.2	2	6.8	15.2
4	23.8	8.5	3	5	14.6
4.5	24.2	8.4	4	3.9	14
5	24	8.2	7	3.3	14.2
5.5	23.5	7.8	9	3	14.6

Specification requirement				
Stability	Flow	VTM	VMA	VFA

>8	3-5	3-5	>13	65-75
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